Cross-Morphology Locomotion via Visual Latent Action World Models

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Background

Locomotion Controllers Are Developed Around a Body

- Legged locomotion controllers are normally expressed using robot-specific variables: *joint positions, torques, motor velocities, and contact measurements*. These variables work well when the robot body is known and remains unchanged.
- However, **when morphology changes, the same numerical command may produce a different physical outcome.**

Robot bodies do not always remain fixed. Leg dimensions can change between prototypes. Payloads and repairs can alter the dynamics. A controller may also need to operate across multiple related platforms.

A locomotion controller maps the robot's current state to a motor command: $\pi : s_t \to a_t$

For the simulated stick insect: $a_t \in \mathbb{R}^{18}$

- The learned locomotion policies commonly use proprioceptive states and body-specific actuation.

The study is called "cross-morphology locomotion transfer" or "cross-embodiment generalization"

$$s_t \xrightarrow{\pi} a_t \xrightarrow[\text{morphology } m]{\text{physical body}} s_{t+1}$$

- So even when morphology is not explicitly given as an input, it is already embedded in the controller through the training process.
- The controller has learned not only "how to walk," but "how this body walks."

Background

Controlled Morphology Test

Does morphology changes alone change the locomotion outcome?

Easiest way to prove morphology gap: Same joint command, different body

A joint command is written in the language of motors. But locomotion behavior happens in the language of contact, support, rhythm, balance, and body motion.

- 3 *Medauroidea extradentata* variants in CoppeliaSim, identical in topology and in their 18-dimensional joint action space, differing only in leg length.
- All three received a identical command sequence.

Property	Short	Medium	Long
Leg-length scale	0.5 x	0.75 x	1.0 x
Number of legs	6	6	6
Controlled joints per leg	3	3	3
Action dimension	18	18	18
Command sequence	Identical	Identical	Identical

If the action space is already shared, should the controller transfer automatically?

The null hypothesis is:

$$H_0: \quad \text{The locomotion outcomes are equivalent across morphologies.}$$

If this hypothesis cannot be rejected, cross-morphology transfer is not yet a meaningful problem in this experimental setting.

Morphology	Path length	Net displacement
Long, 1.0x	5.217 ± 0.069 (cv 1.3%)	4.404 ± 0.187 (cv 4.2%)
Medium, 0.75x	4.149 ± 0.019 (cv 0.4%)	3.569 ± 0.010 (cv 0.3%)
Short, 0.5x	3.228 ± 0.011 (cv 0.3%)	2.729 ± 0.011 (cv 0.4%)

Background

Need a Middle Language of Locomotion Change

We need something more transferable than joint commands, but more precise than a task label.

Motor action

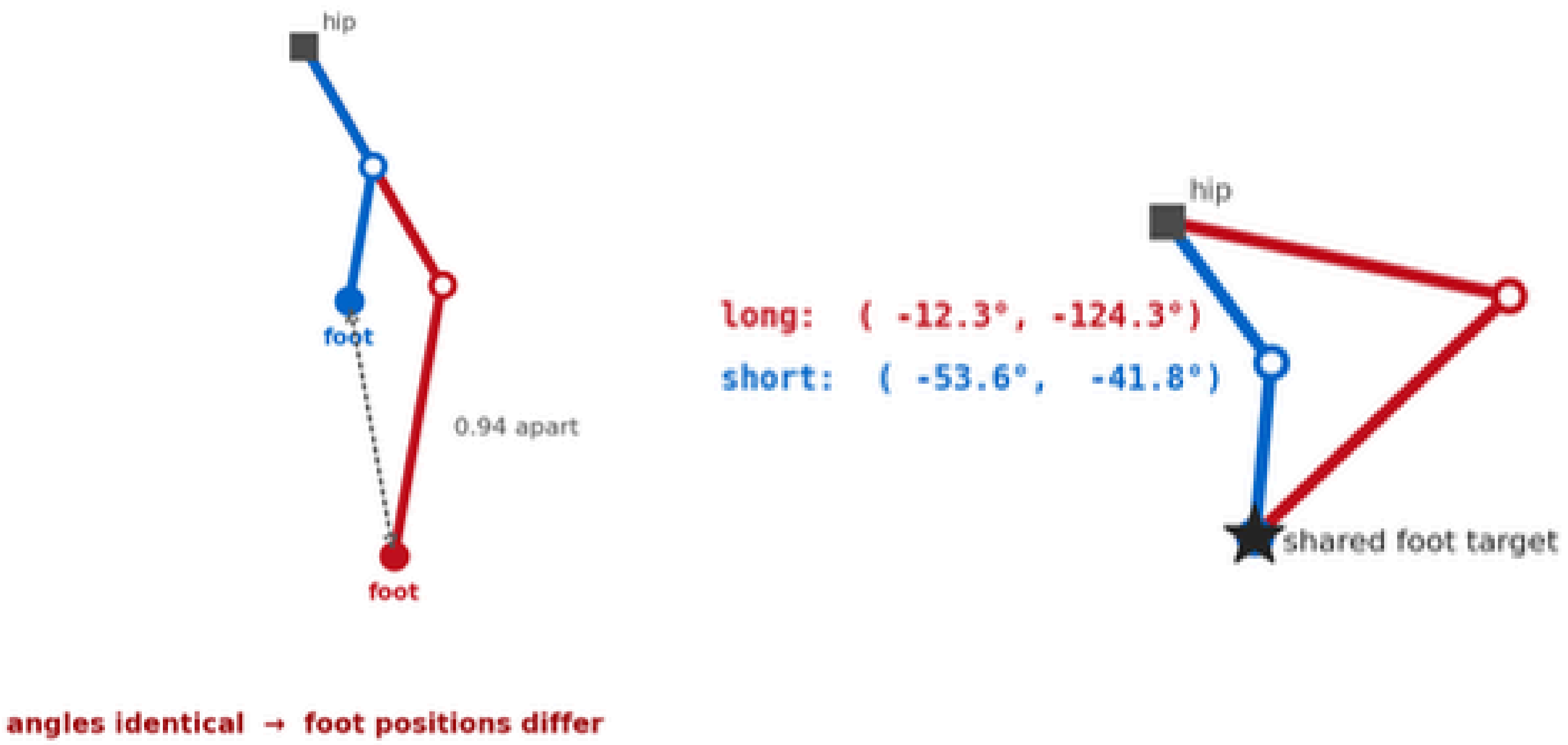
$$a_t = \left[q_{1,t}^{\text{target}}, \ldots, q_{18,t}^{\text{target}} \right]^\top$$

Different bodies may require different motor commands to realize that same behavioral event.

physical locomotion actions

- lifting a foot into swing,
- placing a foot into stance,
- pushing the body forward,
- transferring support between legs,
- correcting lateral instability.

$$\boxed{\text{Same motor command} \neq \text{same physical consequence}}$$



Behavioral grounding	it should describe an observable locomotion change.
Predictive sufficiency	together with the current state, it should help predict what happens next.
Morphology robustness	similar behaviors should remain comparable across different bodies.
Executability	it should retain enough information to recover body-specific motor commands.

- In reinforcement learning, action usually means the numerical output of a policy. ***In locomotion, however, what matters physically is not the number itself.*** What matters is the resulting transition: a foot is lifted, contact is established, the body moves forward, or balance is recovered.
- Therefore, cross-morphology transfer may not require identical motor commands. It may instead ***require preserving the meaning of the locomotion event while allowing its motor realization to change with the body***.

Literature Review

Existing methods transfer by supplying information about the new body

Strategy		Main idea	Body-specific information required	Limitation
Per-body retraining	DreamerV3, Hafner et al. (2023).	Learn a new controller for every morphology	New interactions, reward evaluations, and optimization	Training cost is repeated for each body
Morphology conditioning	QWM, Danesh et al. (2026).	Condition the policy or world model on physical parameters	Leg lengths, masses, torque limits, CAD, URDF, or USD descriptions	Assumes the morphology is accurately known
Online system identification		Infer hidden dynamics from recent interaction history	Proprioception, previous commands, and observed responses	Requires access to the robot's internal signals and sufficient interaction
Shared policy with robot-specific adapters	L3P, Zheng et al. (2025).	Share a common policy backbone while learning separate encoders or decoders	Sensor definitions, joint ordering, actuator conventions, and adaptation data	A new adapter must still be fitted for each platform
Common structure		Although their implementations differ, these approaches obtain information about the new body through at least one of three channels $\boxed{\text{Design specification} \vee \text{Internal sensing} \vee \text{New interaction}}$		

What happens when the body can be observed moving, but cannot be accurately described in advance?

This might occur with animals, undocumented hardware, repaired robots, payload changes, structural wear, or systems whose real physical parameters no longer match their design files.

These methods are effective when the new body can be accurately described or instrumented.
Their common assumption is: The learner has access to an internal description of the body.

Literature Review

Research Gap: The Body Is Observable but Not Described

proprioception does not provide a naturally shared observation space across substantially different bodies.

Existing approaches assume that the new embodiment can be accessed internally

The previous strategies work when at least one of the following is available:

- an accurate geometric and dynamic model,
- known joint and actuator conventions,
- proprioceptive or force measurements,
- permission to collect new interaction data,
- sufficient time to retrain or adapt.

Research gap

Can visual transitions provide a shared behavior-level representation across morphology changes, and how does that representation compare with proprioception?

- For a laboratory robot. CAD, URDF, motor specifications, and onboard sensors is still an approximation.
- The documented body is the set of parameters we believe the system has. The physical body is the system that actually moves and interacts with the environment. In practice, these two descriptions may differ.
- Motivating use cases: Animals or Undocumented machines.
- We may observe how the system walks, while having no access to its joint encoders, actuator commands, or exact dynamic parameters.
- Across substantially different embodiments, proprioceptive vectors may differ in dimension, ordering, units, and semantics. External vision offers a common observation format.

If the body cannot be described directly, perhaps its dynamics can be inferred from how its observable state changes over time.

Literature Review

Why Test Vision When Proprioception Is Available?

Most locomotion systems use proprioception:

- joint angles
- joint velocities
- IMU
- foot forces
- motor states

These are powerful, but they are body-internal and convention-specific.

Vision is different:

- same input format for different bodies
- external observation
- does not require joint conventions or CAD
- captures what happened in the world

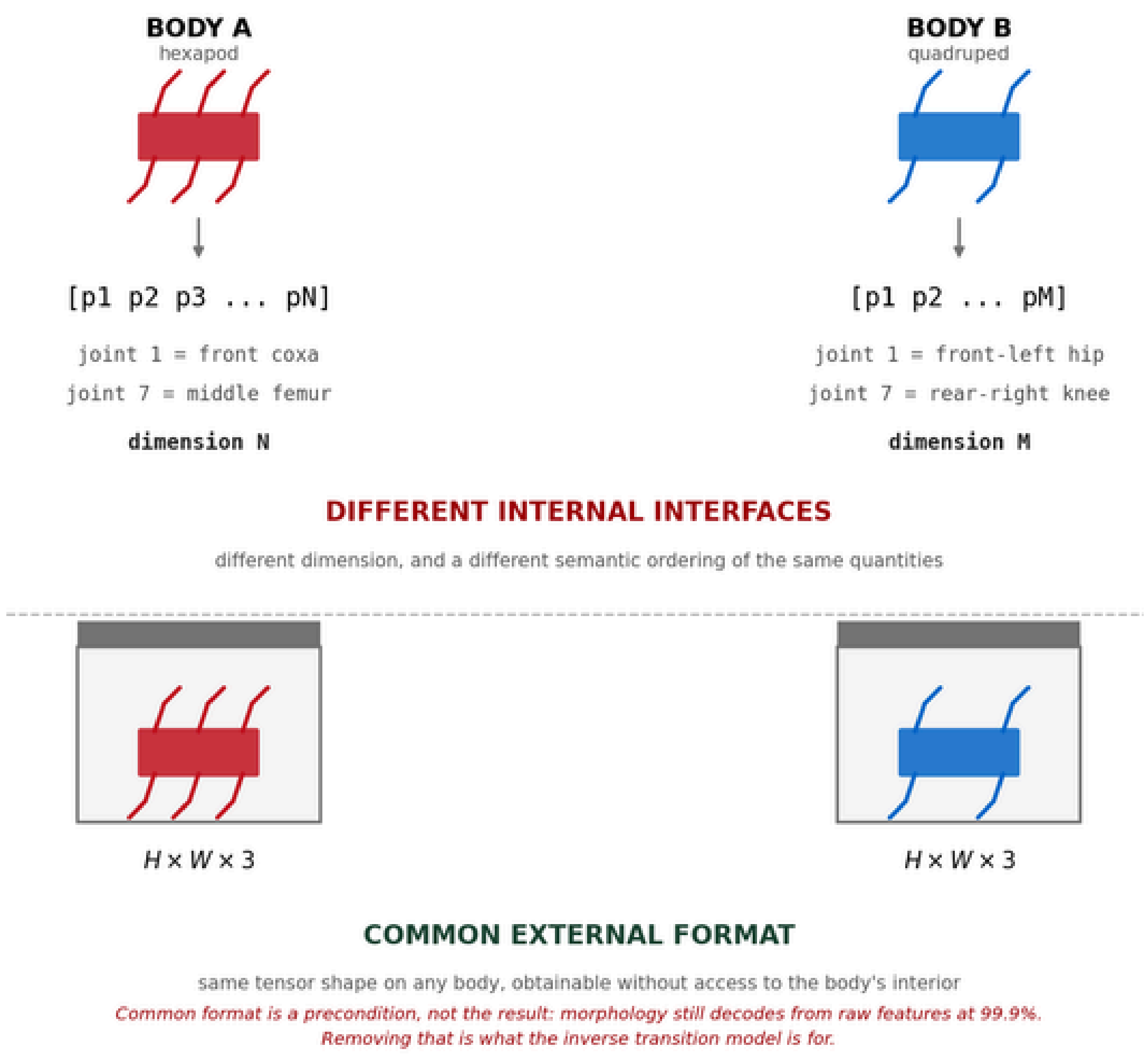
proprioception reports internal command/state
vision reports external consequence

But vision is also dangerous:

A camera sees morphology very clearly.
In the current pilot, raw V-JEPA2 features decode morphology at around 99%.

Can the latent action model extract behavior-relevant change from visual features that strongly contain body shape?

This thesis does not assume vision is superior. It tests whether vision can recover a transferable behavior representation and compares it against proprioception.



Problem Formulation

External observation shows the consequences of hidden body dynamics

Suppose an observer records a sequence:

$$o_1, o_2, o_3, \ldots, o_T$$

where \(o_t\) is an external observation of the body at time \(t\).

A single observation may reveal:

- body shape,
- limb configuration,
- approximate pose,
- foot locations,
- surrounding terrain.

External observation does not give us the robot's exact link lengths, masses, actuator limits, or internal sensor readings. But it gives us something else: ***the visible consequences of those hidden properties.***

A single image mainly tells us what the system looks like. A transition between images tells us how the system behaves.

$$O_t, a_t \rightarrow O_{t+1}$$

These changes reveal:

- which limbs are moving,
- which feet enter or leave contact,
- how the body responds to support,
- whether the body progresses or slips,
- how balance changes over time.

Therefore, the learning problem becomes:

$$s_{t+1} = f_m(s_t, a_t)$$

Given the current observation and some representation of the locomotion action, can we predict the next observation?

The morphology \(m\) may be unknown, but it influences the observable transition:

Can the model learn the relevant dynamics from observed state transitions?

This changes the problem from body description to transition prediction.

Literature Review

Why a World Model?

A world model is often described as a learned simulator used for planning or reinforcement learning. But the important feature for this thesis is that a world model must predict consequences.

A world model learns how actions produce state transitions by approximating the dynamics of the environment

$$\hat{s}_{t+1} = F_\theta(s_t, a_t)$$

For visual observations, images are first converted into compact representations:

$$e_t = E(o_t)$$

The transition model then predicts:

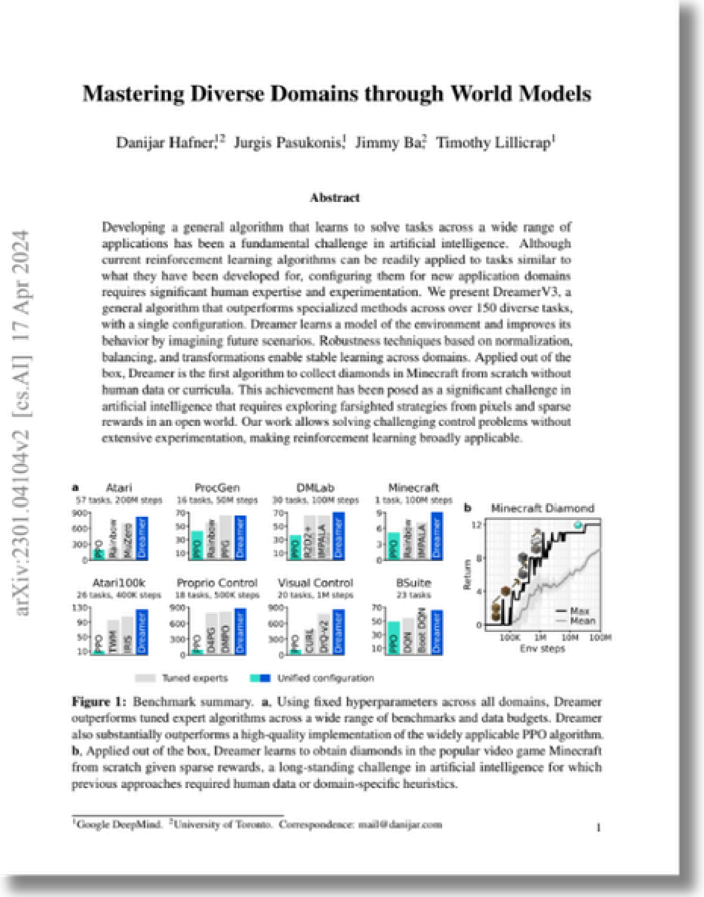
$$\hat{e}_{t+1} = F_\theta(e_t, a_t)$$

The model is trained by comparing the prediction with the actual next observation:

$$\mathcal{L}_{\text{prediction}} = \left\| \hat{e}_{t+1} - e_{t+1} \right\|_2^2$$

The world model is not introduced primarily to generate imagined rollouts. It is introduced because transition prediction provides an objective test of whether an action representation captures meaningful locomotion change.

(Recurrent State Space Model tested with gameplays with explicit action label)



Mastering Diverse Domains through World Models

Danijar Hafner[1,2] Jurgis Pasukonis[1] Jimmy Ba[2] Timothy Lillicrap[1]

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Abstract

Developing a general algorithm that learns to solve tasks across a wide range of applications has been a fundamental challenge in artificial intelligence. Although current reinforcement learning algorithms can be readily applied to tasks similar to what they have been developed for, configuring them for new application domains requires significant human expertise and experimentation. We present DreamerV3, a general algorithm that outperforms specialized methods across over 150 diverse tasks, with a single configuration. Dreamer learns a model of the environment and improves its behavior by imagining future scenarios. Robustness techniques based on normalization, balancing, and transformations enable stable learning across domains. Applied out of the box, Dreamer is the first algorithm to collect diamonds in Minecraft from scratch without human data or curricula. This achievement has been posed as a significant challenge in artificial intelligence that requires exploring farsighted strategies from pixels and sparse rewards in an open world. Our work allows solving challenging control problems without extensive experimentation, making reinforcement learning broadly applicable.

Figure 1: Benchmark summary. **a.** Using fixed hyperparameters across all domains, Dreamer outperforms tuned expert algorithms across a wide range of benchmarks and data budgets. Dreamer also substantially outperforms a high-quality implementation of the widely applicable PPO algorithm. **b.** Applied out of the box, Dreamer learns to obtain diamonds in the popular video game Minecraft from scratch given sparse rewards, a long-standing challenge in artificial intelligence for which previous approaches required human data or domain-specific heuristics.

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Mastering Diverse Domains through World Models (Google DeepMind 2023)

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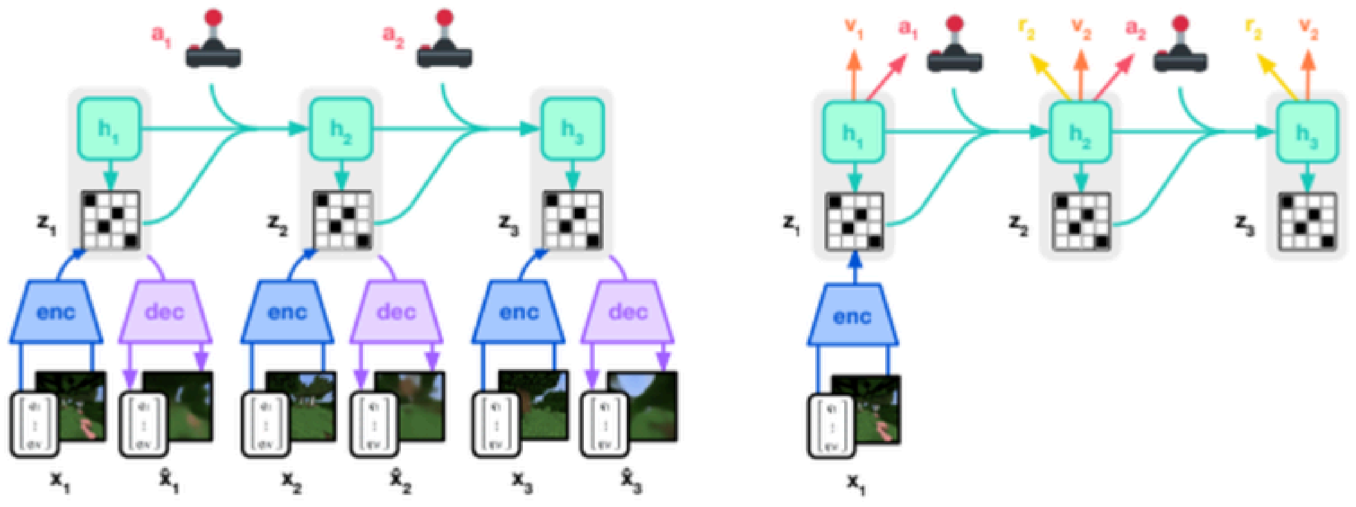


(a) World Model Learning (b) Actor Critic Learning

(Cross-morphology do not share the same action)

DreamerV3 (Hafner et al., 2023) learns a compact latent state and imagines rollouts inside it, so the policy trains without touching the environment. One algorithm with one hyperparameter set covers more than 150 domains. The limitation for this project: each domain still requires its own world model. Nothing carries across bodies, and the method assumes explicit action labels throughout.

A conventional world model still conditions its prediction on the robot's native motor command \(a_t\). But we have already shown that the physical meaning of \(a_t\) changes with morphology. Therefore, a standard world model reproduces the same body-specific action problem inside its transition model.

Literature Review

What Should the World Model Treat as an Action?

The standard formulation is:

$$(e_t, a_t) \rightarrow e_{t+1}$$

But across morphologies:

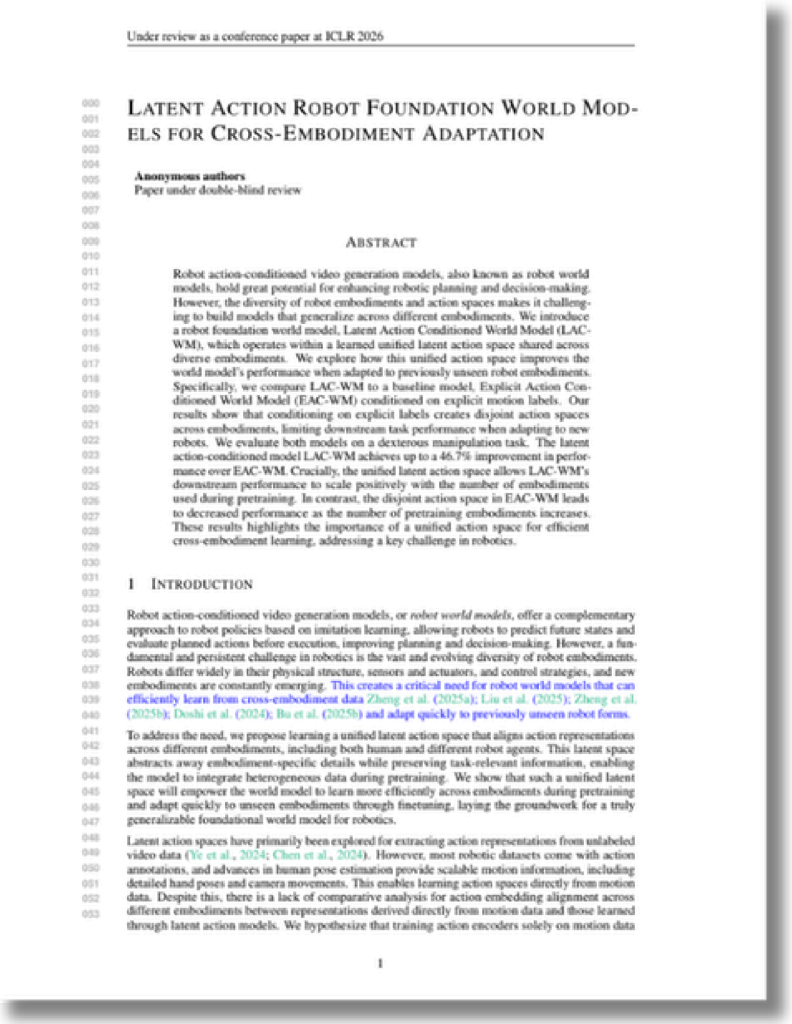
$$a_t = \text{same numerical command}$$

Therefore, instead of assuming that the native joint command is the shared action representation, we ask whether an intermediate variable can be inferred from the observed transition:

$$e_{t+1} - e_t = \text{different physical change}$$

We propose \(z_t\) as a candidate middle language: a compact representation inferred from the locomotion change between consecutive observations.

$$(e_t, e_{t+1}) \rightarrow z_t$$



Under review as a conference paper at ICLR 2026

LATENT ACTION ROBOT FOUNDATION WORLD MODELS FOR CROSS-EMBODIMENT ADAPTATION

Anonymous authors
Paper under double-blind review

ABSTRACT

Robot action-conditioned video generation models, also known as robot world models, hold great potential for enhancing robotic planning and decision-making. However, the diversity of robot embodiments and action spaces makes it challenging to build models that generalize across different embodiments. We introduce a robot foundation world model, Latent Action Conditioned World Model (LAC-WM), which operates within a learned unified latent action space shared across diverse embodiments. We explore how this unified action space improves the world model's performance when adapted to previously unseen robot embodiments. Specifically, we compare LAC-WM to a baseline model, Explicit Action Conditioned World Model (EAC-WM) conditioned on explicit motion labels. Our results show that conditioning on explicit labels creates disjoint action spaces across embodiments, limiting downstream task performance when adapting to new robots. We evaluate both models on a dexterous manipulation task. The latent action-conditioned model LAC-WM achieves up to a 46.7% improvement in performance over EAC-WM. Crucially, the unified latent action space allows LAC-WM's downstream performance to scale positively with the number of embodiments used during pretraining. In contrast, the disjoint action space in EAC-WM leads to decreased performance as the number of pretraining embodiments increases. These results highlights the importance of a unified action space for efficient cross-embodiment learning, addressing a key challenge in robotics.

1 INTRODUCTION

Robot action-conditioned video generation models, or *robot world models*, offer a complementary approach to robot policies based on imitation learning, allowing robots to predict future states and evaluate planned actions before execution, improving planning and decision-making. However, a fundamental and persistent challenge in robotics is the vast and evolving diversity of robot embodiments. Robots differ widely in their physical structure, sensors and actuators, and control strategies, and new embodiments are constantly emerging. This creates a critical need for robot world models that can efficiently learn from cross-embodiment data Zheng et al. (2025a); Liu et al. (2025); Zheng et al. (2025b); Doshi et al. (2024); Bu et al. (2025b) and adapt quickly to previously unseen robot forms.

To address the need, we propose learning a unified latent action space that aligns action representations across different embodiments, including both human and different robot agents. This latent space abstracts away embodiment-specific details while preserving task-relevant information, enabling the model to integrate heterogeneous data during pretraining. We show that such a unified latent space will empower the world model to learn more efficiently across embodiments during pretraining and adapt quickly to unseen embodiments through finetuning, laying the groundwork for a truly generalizable foundational world model for robotics.

Latent action spaces have primarily been explored for extracting action representations from unlabeled video data (Ye et al., 2024; Chen et al., 2024). However, most robotic datasets come with action annotations, and advances in human pose estimation provide scalable motion information, including detailed hand poses and camera movements. This enables learning action spaces directly from motion data. Despite this, there is a lack of comparative analysis for action embedding alignment across different embodiments between representations derived directly from motion data and those learned through latent action models. We hypothesize that training action encoders solely on motion data

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LATENT ACTION ROBOT FOUNDATION WORLD MODELS FOR CROSS-EMBODIMENT ADAPTATION

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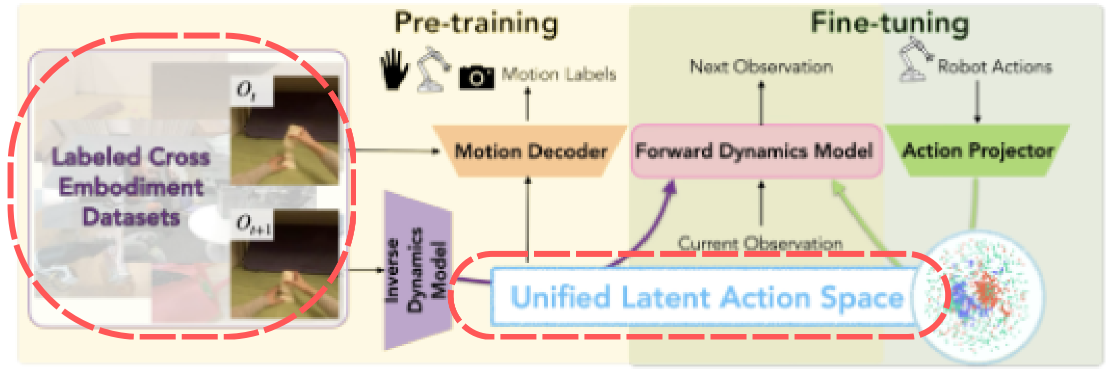


LAC-WM (Huang et al., ICML 2026) discards explicit action labels as the conditioning signal. An inverse model infers an abstract action z from consecutive observations, and the world model is conditioned on z rather than on any robot's native command vector. One latent space then covers several embodiments, and adding embodiments improves it rather than fragmenting it.

- LAC-WM problem: different robots have different action formats
- This project: robots share the same 18D action format,
  but the same 18D command has different dynamics meaning

Research Question

Can observable locomotion change become a shared action representation?

Primary research question

Can a latent action inferred from visual state transitions preserve behavior-relevant locomotion information while reducing morphology-specific information across different leg lengths?

Secondary question

How does the visual latent representation compare with raw visual features, native joint commands, and proprioceptive observations?

Testable hypotheses	
H1 : Behavioral information	The latent action should preserve information about locomotion events such as foot contact and support transition:
H2: Reduced morphology dependence	The morphology should be less recoverable from z_t than from the raw visual representation:
H3: Predictive isefulness	The current state and latent action should predict the next state:
H4: Added value over native commands	A latent-conditioned dynamics model should be compared against a model conditioned directly on the 18D joint command. (If the native-command model performs equally well across morphologies, the latent action may be unnecessary in this controlled setting.)

***Success is not defined as "vision wins"**
The intended result is a scientific comparison: Determine which information source produces the most transferable behavior representation, under which assumptions, and at what cost.

Experimental Design

Can observable locomotion change become a shared action representation?

Execution Plan

Platform:
- **CoppeliaSim**

Robot:
- Medauroidea extradentata stick-insect model
- 3 morphologies (short / medium / long)
- each with 6 legs, 3 controlled joints/leg

Native action:
- joint position targets ℝ^18

Data collection policy: TBD

Behaviors:
- walk (might extent to turn / stop)

Train morphologies:
- short + long leg

Medium leg is an interpolation test set
(Extrapolation future direction)

Test morphology:
- medium leg (Step 2 transfer test)

Camera Control
- Script-created camera identical across morphologies.
- Third-person view.
- Approximately 40° elevation.
- 45° field of view.
- Matte, lightly textured floor.
- Camera follows the body from a fixed relative offset.
- No empty background pixels in the validated configuration.

Validation Logic

Step 0: Does a morphology gap exist?

Step 1: Does frozen vision contain locomotion information?

Step 2: Does z_t preserve behavior while reducing morphology information?

Step 3: Does z_t improve held-out transfer over raw commands?

what model

what is recorded

RGB	Proprioception	Command	Contact	Next RGB

Step 0 → Step 1 → Step 2 → Step 3

Evaluation

What Counts as Success?

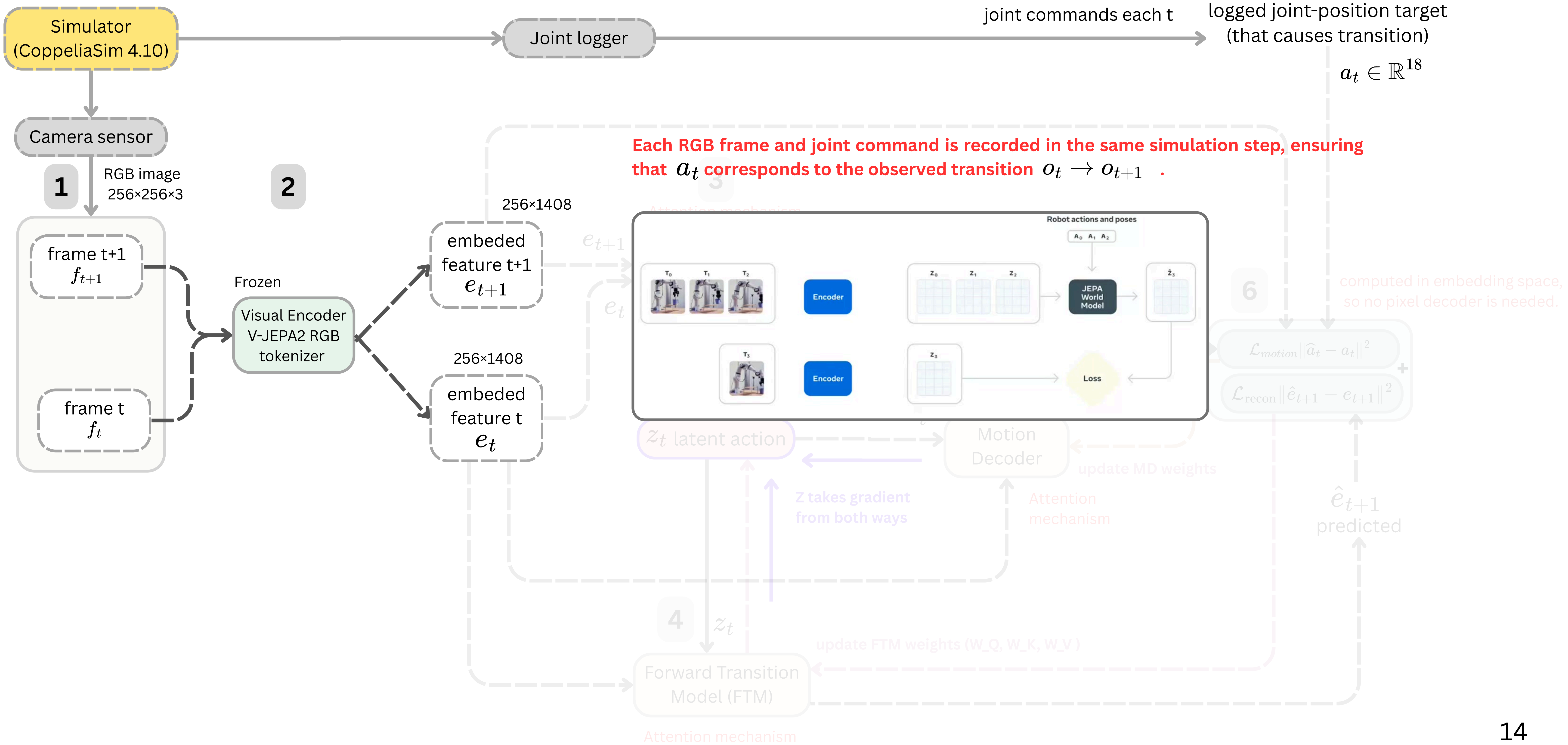
Requirement	Measurement	Success criterion
Behavior preservation	Foot-contact balanced accuracy and macro-F1	\(z_t\) transfers better than raw visual features \(e_t\)
Morphology robustness	Morphology probe accuracy; silhouette as supporting evidence	Morphology is less decodable from \(z_t\) than from \(e_t\)
Predictive value	Next-embedding error on held-out medium	\(F(e_t,z_t)\) transfers better than or degrades less than \(F(e_t,a_t)\)
Executability	IK command reconstruction MAE	\(MD(e_t,z_t)\) outperforms \(MD(e_t,0)\)
Adaptation efficiency	Medium-leg learning curve	Pretrained model reaches the target error using fewer episodes than training from scratch

$$\mathrm{BehaviorTransfer}(z_t) > \mathrm{BehaviorTransfer}(e_t)$$
$$\mathrm{MorphologyDecode}(z_t) < \mathrm{MorphologyDecode}(e_t)$$

Both conditions must hold simultaneously. Reducing morphology information alone is insufficient because a collapsed representation would also make morphology difficult to decode.

Pipeline (Focused)

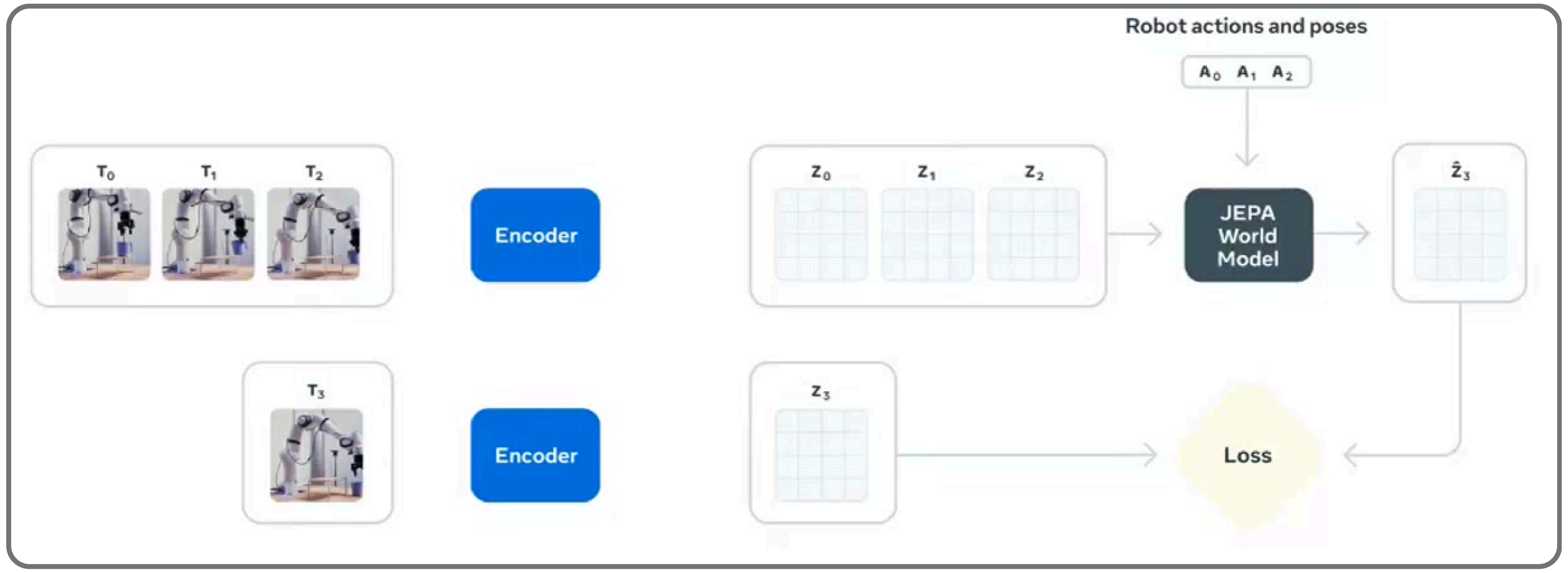
Data source and Pre-processing



Implemented Front-End

V-JEPA2

V-JEPA2 is self-supervised on roughly one million hours of video, with an objective that predicts masked content in representation space rather than pixel space. That objective rewards motion-relevant features, which is why it is a reasonable starting point for gait.



https://ai.meta.com/research/vjepa/

We adopt V-JEPA 2's frozen RGB tokenizer as our visual encoder.

- extracting per-frame embeddings e_t ∈ ℝ^{256×1408} from simulation video without any fine-tuning.
- These embeddings serve as input to our ITM and FTM, which learn a morphology-agnostic latent action space z_t shared across stick insect morphologies.
- V-JEPA2 is a reasonable starting point because it was pretrained on large-scale video; whether its features contain locomotion information is tested empirically in Step 0

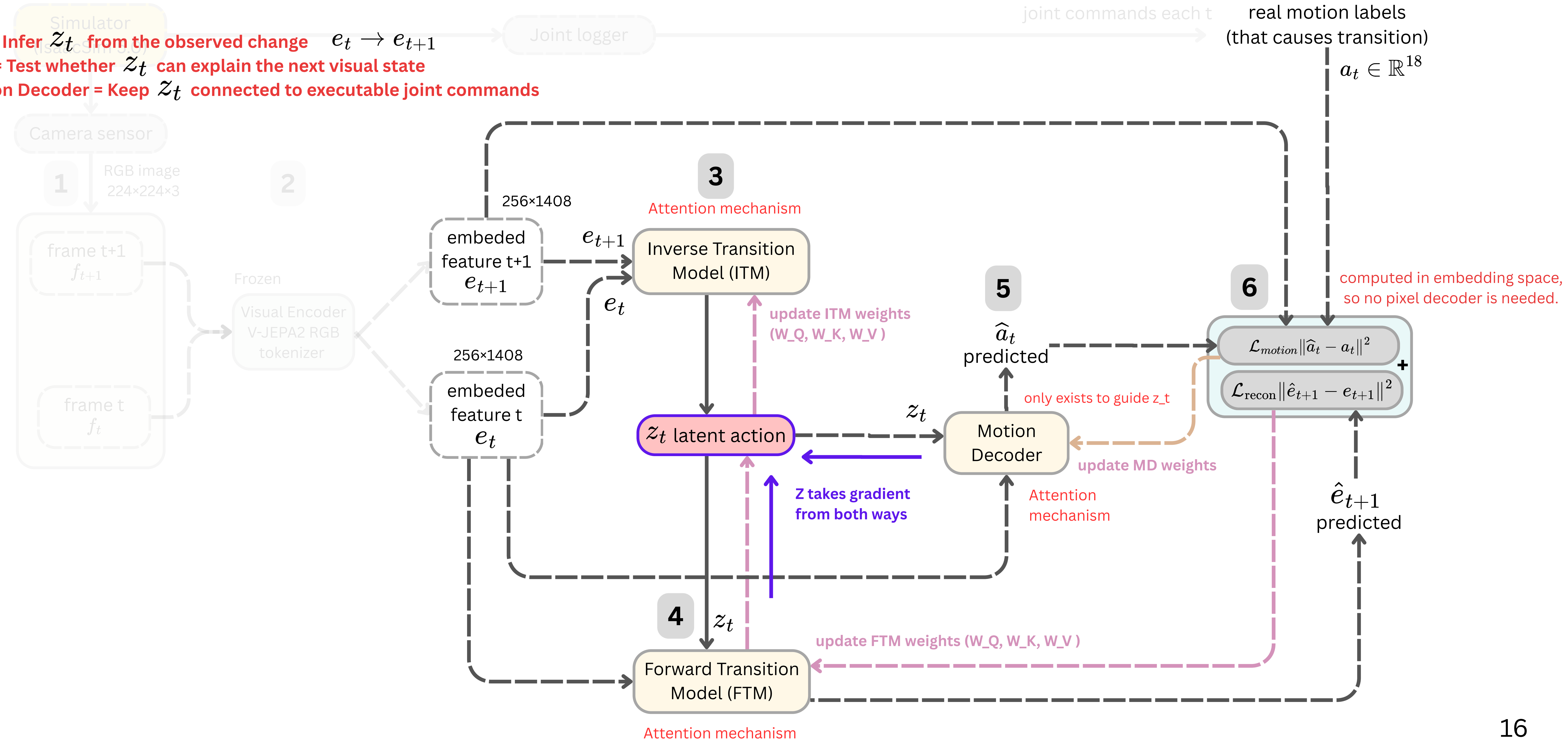
frame_t ∈ ℝ^{256×256×3} → **Patch Splitting**: Split into 16×16 pixel patches 256×256 image ÷ 16 = 16×16 grid → 256 patches → **Positional Embedding**: add position info to each token (so model knows patch [0,0] vs patch [15,15]) → **ViT Transformer (frozen))**: 256 tokens attend to each other each token becomes context-aware (patch of leg knows about patch of ground next to it) → e_t ∈ ℝ^{256×1408}

trained weight from 1 million hours of internet video + 1 million images

Pipeline (Focused)

How is z_t learned

ITM = Infer z_t from the observed change $e_t \rightarrow e_{t+1}$
FTM = Test whether z_t can explain the next visual state
Motion Decoder = Keep z_t connected to executable joint commands



Pipeline (Focused)

Proposed Trainable Modules

ITM (Inverse State-Transition Model) $(e_t, e_{t+1}) \to z_t$

Compresses the observed transition into a latent action that explains what changed between the two states.

input: [e_t¹, e_{t+1}¹] ∈ ℝ^{512×1408} + q_t ∈ ℝ^{512}
- × 4 Causal Self-Attention → e_{t+1} absorbs context of e_t
- × N Cross-Attention → q_t queries S, focuses on moving patches

Output: z_t ∈ ℝ^{64} (latent action)

z_t latent action

FTM (Forward State-Transition Model) $(e_t, z_t) \to \hat{e}_{t+1}$

input: e_t² ∈ ℝ^{256×1408} + z_t ∈ ℝ^{64}
- × 8 Transformer Blocks: self-attn(e_t²) → self-attn(z_t) → cross-attn(e_t² queries z_t)

Output: ê_{t+1} ∈ ℝ^{256×1408}
Loss: L_recon = ||ê_{t+1} − e_{t+1}¹||²

Tests whether the current state and z_t are sufficient to explain the next visual state.

Motion Decoder $(e_t, z_t) \to \hat{a}_t$

Input: z_t ∈ ℝ^{64} + e_t¹ ∈ ℝ^{256×1408}
- cross-attn(z_t queries e_t¹) + MLP

Output: â_t ∈ ℝ^{18}
Loss: L_motion = ||â_t − a_t||²
(drop after pretraining finish)

Requires z_t to retain information connected to executable, body-specific joint commands.

L_total = L_recon + λ · L_motion

ITM receives training signals from both objectives through z_t.

Pipeline (Focused)

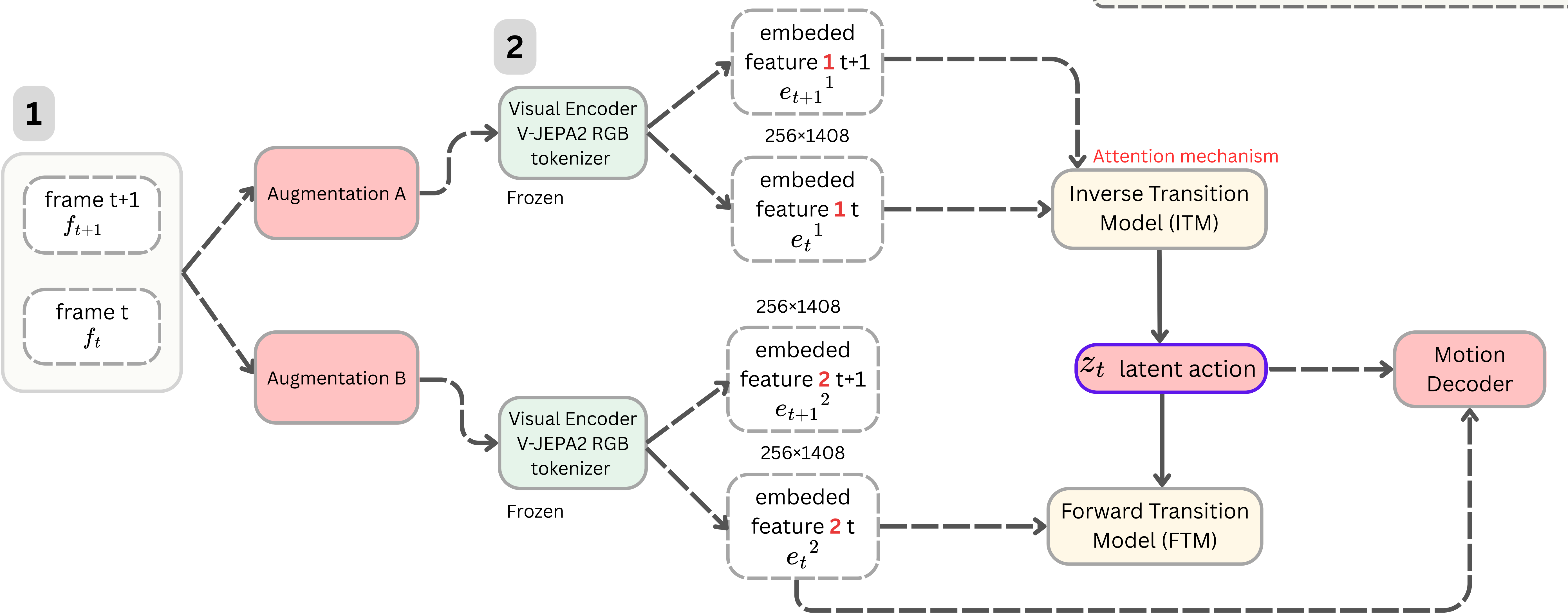
What prevents a shortcut?

In the previous pipeline, z_t may become a compressed copy of the next state rather than a representation of the action between states.

Fix

Use the same augmentation parameters for f_t and f_{t+1} inside each branch, so the temporal change is preserved. However, Augmentation A and B must be independently sampled.

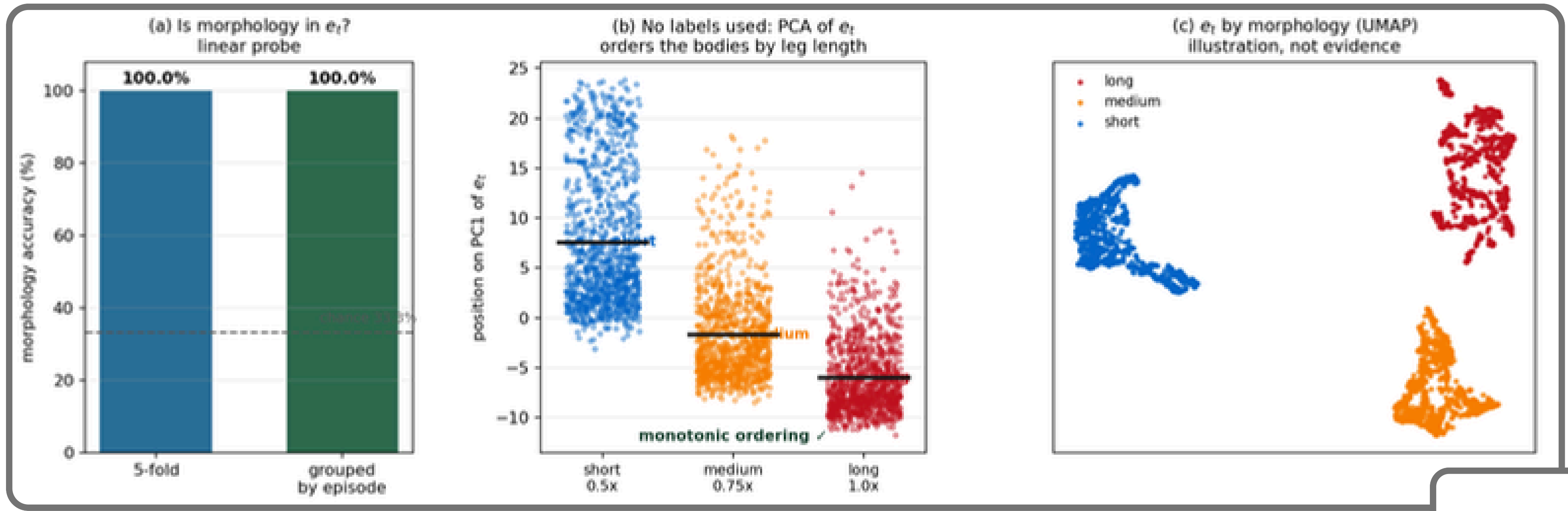
The reason for using 2 views is that the ITM sees Augmentation A, while the FTM must predict Augmentation B. This makes it harder for the ITM to copy the exact future embedding e_{t+1} into z_t



Preliminary Check

Frozen V-JEPA2 Features Encode Morphology/Session Identity

Experimental Question: Does the frozen visual representation e_t contain locomotion information, and how strongly does it contain morphology/session identity?
Input: frozen V-JEPA2 e_t (1408-d per frame (ViT-g/16, mean-pooled over 256 patches, encoder never trained))
Model: logistic regression (linear probe) + standardization
Data: 3 bodies × 5 episodes × 200 steps = 3000 frames, same bit-identical command sequence
Evaluation: 5-fold cross-validation



Morphology Encoding

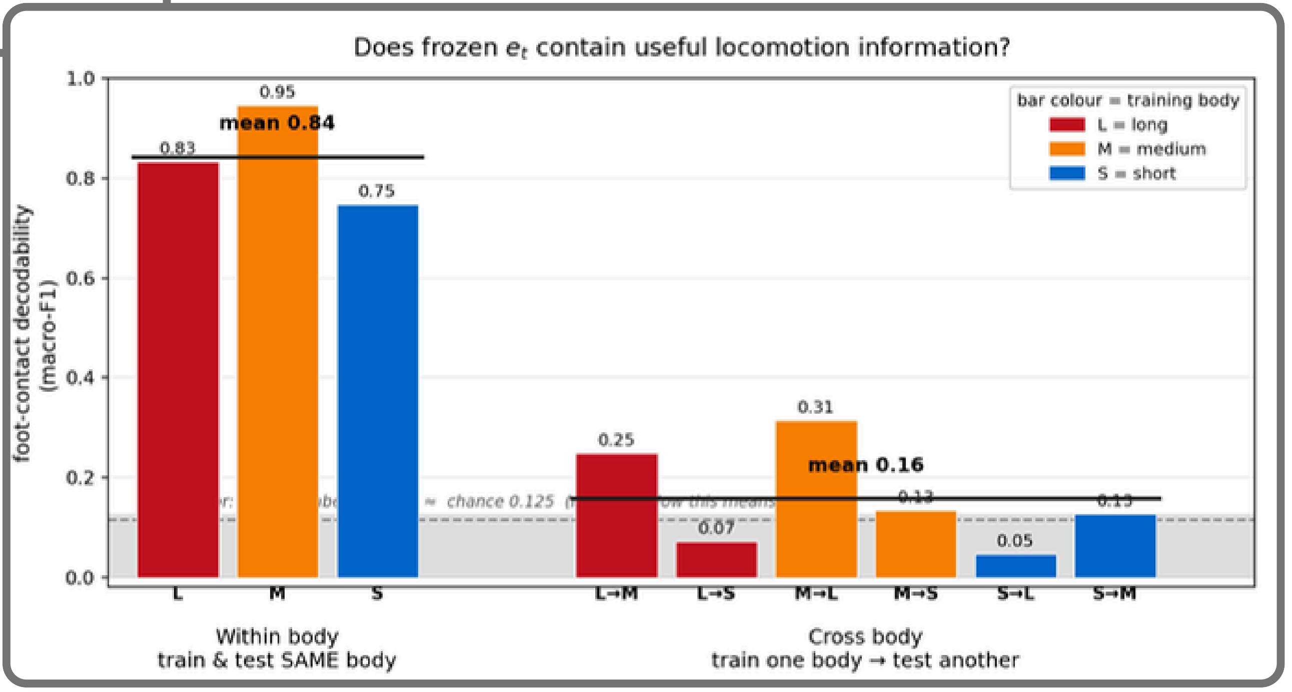
Target: which body (long / medium / short) — 3 classes, chance 33%
Three tests, weakest → strongest:
(a) supervised probe → ~100%, holds under episode-grouped CV (not frame memorisation)
(b) unsupervised PCA → PC1 orders short < medium < long with no labels (self-organises)
(c) UMAP → illustration only, not evidence

Reads as: body identity is a dominant axis of e_t.
This is the baseline z_t must reduce.

Behavior Encoding

Target: which feet are planted (foot-contact pattern, top-8) — macro-F1, chance 0.125
Three splits:
within-body train & test same body → 0.84 (contact is decodable)
cross-body train one body, test another → 0.16 (does not transfer)
shuffle permuted labels → ≈ chance (confirms 0.84 is real, not an artefact)

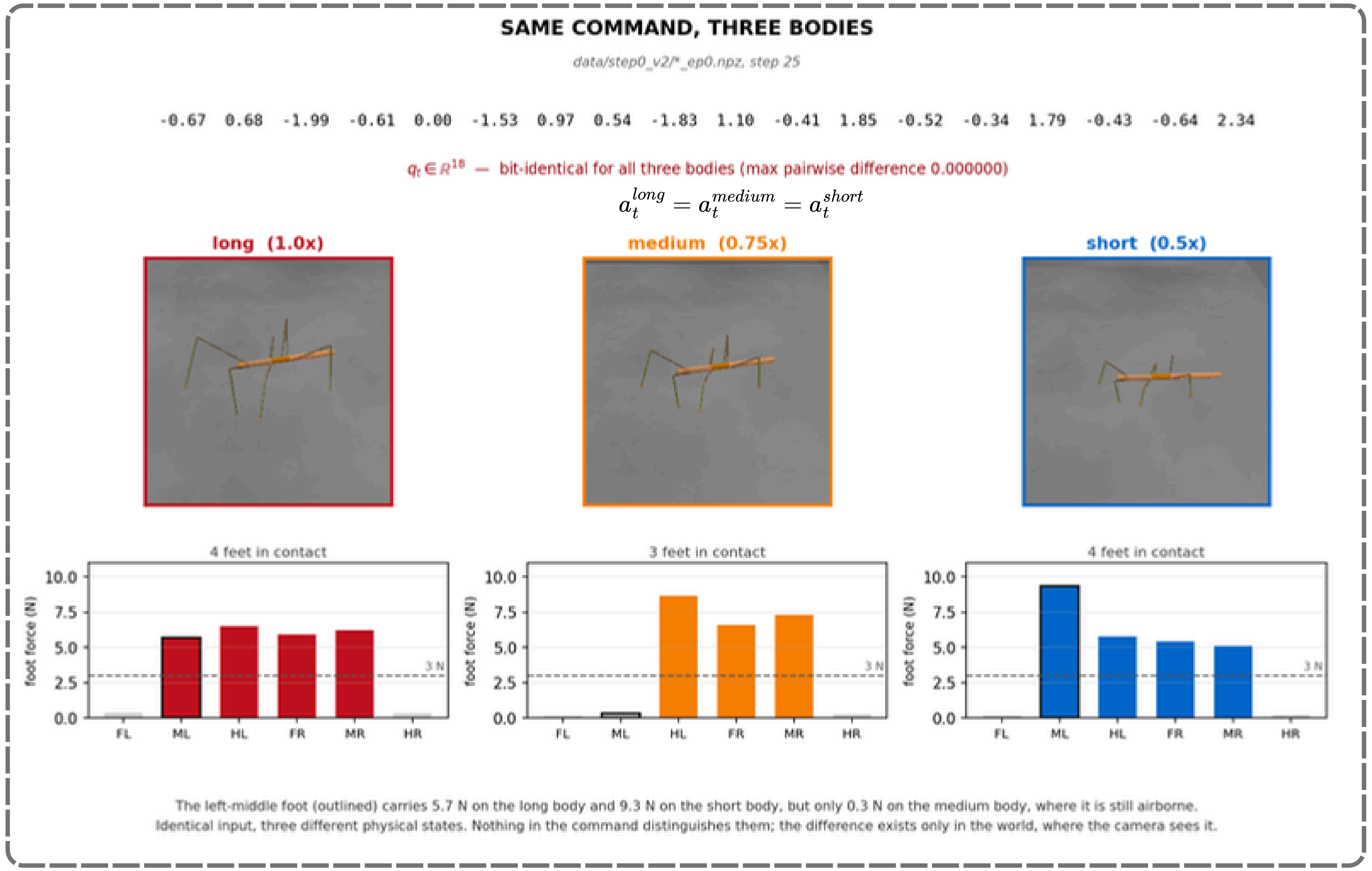
Reads as: e_t encodes behavior, but entangled with body shape (the wider the leg-length gap, the worse it transfers (L↔S worst).
This is the signal z_t must keep while making it body-independent.



Preliminary Check

Pilot Limitation: Identical Commands, Different Physical States

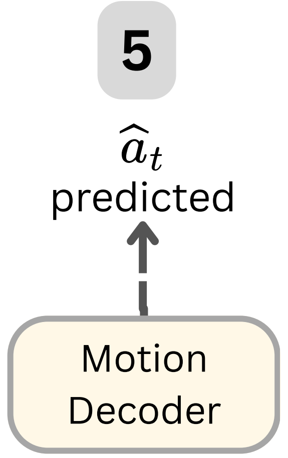
Identical commands produce different contact states. The difference is absent from the command but present in the resulting visual and proprioceptive state.



Type: Direct measurement, reads the simulator state at one timestep
Controlled: the 18D joint command q_t — bit-identical across all 3 bodies
Measured: per-foot contact force (N) from the sim's force sensors, and the rendered RGB frame
Varied: only leg length (1.0 / 0.75 / 0.5×)
Frame shown: long/medium/short_ep0, step 25

Same command → different physical state:
left-middle foot carries 5.7 N (long) / 9.3 N (short) but only 0.3 N (medium) — airborne on the medium body while planted on the other two.

The command cannot tell the bodies apart — it is identical, yet the outcome is not. This is why the shared a_t makes the latent action vacuous, and why per-body commands (IK retargeting) are needed before training.

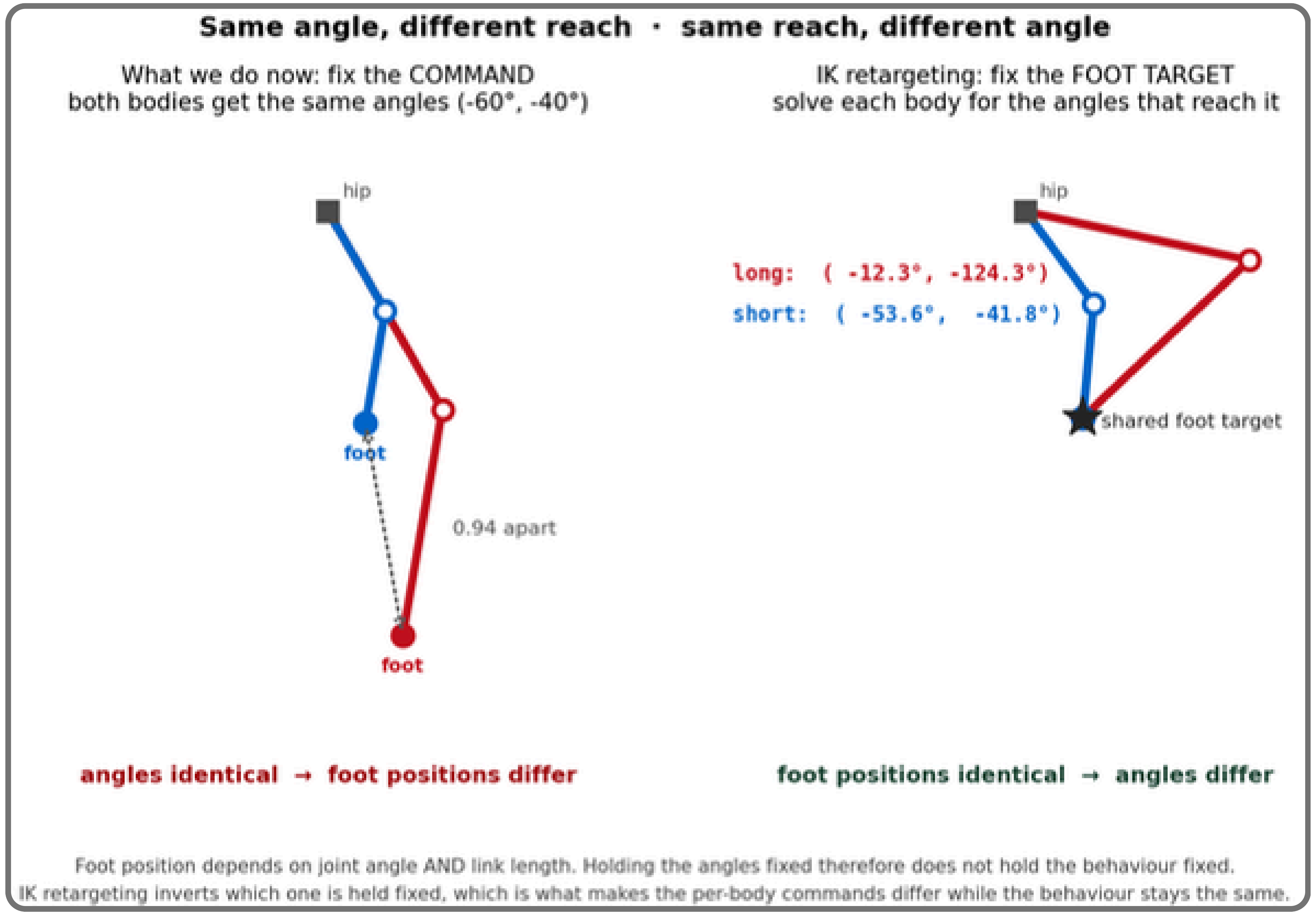


The Motion Decoder cannot demonstrate morphology-specific command recovery !

Preliminary Check

Better Expert Data: Shared Targets, Body-Specific Commands

shared task-space target → body-specific joint commands



Type: Concept forward/inverse kinematics
Bodies illustrated: link length 1.0× and 0.5×

Foot position depends on angle AND link length. Fixing angles does not fix behaviour; fixing the foot target does — and the per-body angle difference is exactly the body-specific command the Motion Decoder must learn to produce.

- old CSV = joint angles + contact only
  Problem: foot trajectory must be derived via forward kinematics).
- new 66k CSV = foot trajectories logged directly
  Provide: ready-made IK targets, plus sim binary contact.

Final Experimental Protocol

Final Experimental Protocol: Train on Two Bodies, Test on One

Execution Plan

Platform:
- **CoppeliaSim**

Robot:
- Medauroidea extradentata stick-insect model
- 3 morphologies (short / medium / long)
- each with 6 legs, 3 controlled joints/leg

Native action:
- **task-space foot trajectories**

Data collection policy: **IK-retargeted expert dataset**

Behaviors:
- walk (might extent to turn / stop)

Train morphologies:
- short + long leg

Test morphology:
- medium leg (Step 2 transfer test)

Camera Control

- Script-created camera identical across morphologies.
- Third-person view.
- Approximately 40° elevation.
- 45° field of view.
- Matte, lightly textured floor.
- Camera follows the body from a fixed relative offset.
- No empty background pixels in the validated configuration.

Validation Logic

- ✓ Step 0: Does a morphology gap exist?
- ✓ Step 1: Does frozen vision contain locomotion information?
- Step 2: Does z_t preserve behavior while reducing morphology information?
- Step 3: Does z_t improve held-out transfer over raw commands?

what model

what is recorded

RGB	Proprioception	Command	Contact	Next RGB

Step 0 → Step 1 → Step 2 → Step 3

Decisive Ablation

Does Latent Action Add Value Beyond Raw Joint Commands?

Compare models under the same encoder, FTM capacity, dataset, and training budget:

Transition model	Conditioning input	What it tests
Observation only	\(F(e_t,0)\)	How predictable is motion without an action?
Raw action	\(F(e_t,a_t)\)	Standard body-specific command representation
Latent action	\(F(e_t,z_t)\)	Proposed transition-based representation

Primary comparison $\mathcal{L}_{pred}^{medium}(e_t, z_t) < \mathcal{L}_{pred}^{medium}(e_t, a_t)$

The latent representation is useful only if it improves held-out transition prediction or adaptation relative to raw commands.

Possible Outcomes

What Would the Results Mean?

Behavior in \(z_t\)	Morphology in \(z_t\)	Prediction/transfer	Interpretation
Preserved	Reduced	Improved	Intended morphology-
Preserved	Still high	Improved	Useful representation,
Lost	Reduced	Poor	Representation collapse or
Preserved	Reduced	No improvement	Clean latent space, but no
Preserved	Reduced	Worse than \(a_t\)	Raw commands remain more

Success requires all three:

$$\text{behavior preserved} \quad + \quad \text{morphology reduced} \quad + \quad \text{transfer improved}$$

Scope and Limitations

Scope and Limitations

Included
- Simulation-based hexapod locomotion
- Three leg-length morphologies
- Fixed third-person RGB camera
- Forward locomotion and foot-contact behavior
- Medium-leg interpolation test
- ITM, FTM, Motion Decoder, and frozen V-JEPA2 encoder

Not claimed
- Real-robot or animal transfer
- Extrapolation beyond the training morphology range
- Camera-viewpoint invariance
- Generalization to manipulation or complex terrain
- Fully autonomous control through z_t

IK creates comparable task-space objectives, but it does not guarantee identical contact dynamics or identical behavior across morphologies.

Contributions and Milestones

Expected Contributions and Research Milestones

Expected contributions

- A controlled benchmark for studying locomotion representation across changing body morphology.
- A visual latent-action world model combining ITM, FTM, and command reconstruction.
- An evaluation framework separating behavior preservation, morphology leakage, predictive sufficiency, executability, and adaptation efficiency.
- Evidence establishing whether visual latent actions improve held-out morphology transfer over raw joint commands.

Objectives

- To design and train a Latent Action Space pipeline using Inverse State-Transition Model (ITM), Forward State-Transition Model (FTM), and Motion Decoder on video data and Action Labels automatically logged by simulation, from a Stick Insect robot across 3 morphologies.
- To prove that the learned Latent Action is Morphology-Agnostic by using Principal Component Analysis (PCA) to analyze whether z_t from robots of different morphologies performing the same behavior cluster together.
- To test knowledge transfer to an unseen morphology (Medium Leg) not seen during Pretraining, and ***measure whether the resulting World Model reduces the amount of data required for a new morphology***.

Stage	Completion criterion
Dataset redesign	Valid IK gait, synchronized observations, distinct per-body commands
Representation training	Stable ITM–FTM–MD optimization without collapse
Representation evaluation	Behavior retained while morphology/session signal decreases
Transfer evaluation	Held-out medium results compared with required baselines
Thesis conclusion	Determine where latent action helps, fails, or requires stronger supervision

The contribution is not an assumption that latent action must work, but a controlled test of whether transition-based visual representations provide measurable cross-morphology value.